

1 Reuse Symbolic Execution

V : location set.

$Expr$: concrete values or symbolic expression.

$s \in \mathbb{S} = (i, \sigma, \pi)$: state is a tuple(next statement, store, path condition), $\sigma := V \rightarrow Expr$

$$\text{REUSESYMEXPR} \frac{\begin{array}{l} s_0 : (i, \sigma_0, \pi_0), s_1 : (i, \sigma_1, \pi_1) \in \mathbb{S} \\ s_0 \neq s_1 \quad U(s_0)(i) = U(s_1)(i) \\ \forall u \in U(s_0)(s_1). \sigma_0(u) = \sigma_1(u) \end{array}}{\forall d \in D(s_0)(i) \cap D(s_1)(i). \sigma_0(d) = \sigma_1(d)}$$

Vanilla symbolic execution: $\forall s \in \mathbb{S}, f_i(s)$

Sparse symbolic execution

$v@i$ means variable v defined at statement i

data dependence: (p, q)

$$\text{REUSESYMEXPR} \frac{\begin{array}{l} s : (i, \sigma, \pi) \in \mathbb{S}, (p@i, q@i') \in \mathcal{E}_d \\ s_0 : (i', \sigma_0, \pi_0), s_1 : (i', \sigma_1, \pi_1) \in \text{child}(s) \end{array}}{f_{i'}(s_0)|_q = f_{i'}(s_1)|_q}$$

over approximate defined and used variables?

How to abstract state and partition?